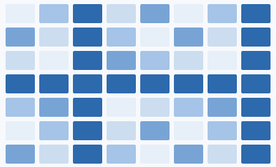
FastVCL Sparse-to-tube video copy localization

PROPOSED METHOD · SCHEMATIC

1 Fixed features

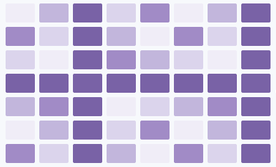
Query A / reference B

Query A



F_A: M × 256

Reference B



F_B: N × 256

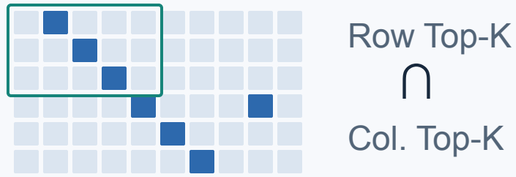
L2 normalization
+ validity masks

2 Sparse matching

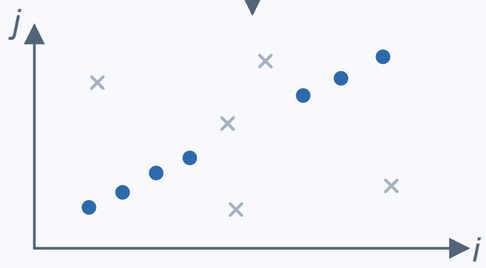
Sample → compare → retain



Sampling plans g ∈ G



Tiled cosine · mutual Top-K



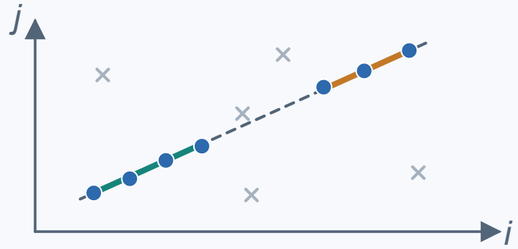
Deduplicate coordinates; cap

Anchor set {(i, j, S_{ij})}

3 Temporal proposals

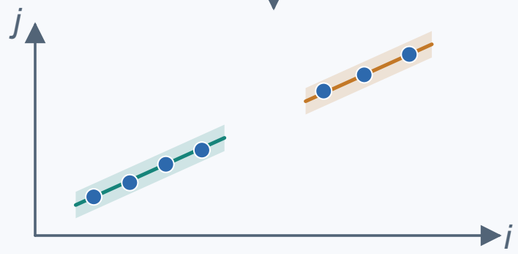
Fit → validate → split

RANSAC: $j \approx \alpha_r i + \beta_r$



Distinct-time support on A and B

Split long gaps → refit / validate



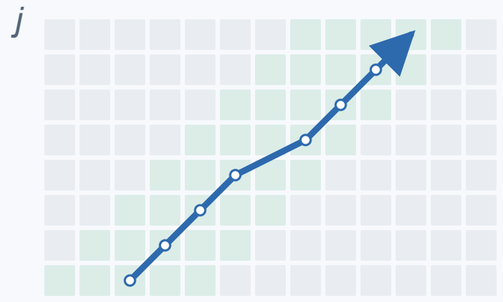
Extend A: e_A B band: ±w_r

Finite T_r: $\ell_r \le i < u_r$

4 Refinement & output

Align → recover → deduplicate

Cosine scores only inside T_r



Local DP with a zero restart

$H_{ij} = \max\{0, r_{ij}, \text{continuation}\}$

Traceback → trim → score

↻ Block matched cells; repeat



Score q · min(IoU_A, IoU_B) dedup.

Full-resolution normalized descriptors + validity masks

● anchor — track ▭ tube → local path

Output: (a_s, b_s, a_e, b_e, q) · feature-index coordinates